Bugs found: 5

1. Missing null check after malloc: the return value of malloc(20) is never verified for NULL before use.

2. Use-after-free: memory pointed to by 'dangling' is written to after being freed with free(dangling).

3. Undefined behavior from integer overflow: '1 << 40' shifts a 32-bit int by 40 bits, exceeding its width.

4. Off-by-one buffer overflow: the loop condition 'i <= 20' allows writing to buf[20], one byte past the 20-byte allocation.

5. Memory leak: the buffer allocated with malloc(20) and assigned to 'buf' is never freed before return.